

2023



Redfin Uses Amazon SES to Reliably Deliver Millions of Property Listing Emails Every Day

Redfin relies on Amazon SES to deliver millions of daily property listing emails to customers, get accurate bounce and complaint notifications, and

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20+ million

1,000 emails per second



2

Accurate bounce and complaint notifications

improves email feedback handling system and email reputation

Overview

Redfin offers brokerage, rentals, lending, title insurance, and renovation services to customers in the US and Canada. To improve email deliverability metrics, cut costs, and save engineering time, Redfin migrated its email environment to [Amazon Simple Email Service](#) (Amazon SES).

Using Amazon SES, Redfin delivers millions of customer emails every day, provides accurate and reliable bounce and complaint notifications, reduces engineering costs, and gives engineers more time to focus on business growth.



Opportunity | Exploring an Email Migration to the Cloud

[Redfin](#) is a technology-powered real estate company that helps people find a place to live through brokerage, rentals, lending, title insurance, and renovation services. The organization operates in more than 100 markets in the US and Canada.

Email plays a major role in the company's user experience. Redfin sends millions of emails each day, which drive traffic to the Redfin website and brokerage services. The emails notify customers when a new home that matches their search criteria hits the market or drops its price and also recommend homes that are like [st.](#) "We have a variety of business objectives with these email notifications," says Rohit Misar, Director of Engineering for Redfin. "We are growing our audience by providing rental listing notifications in addition to for-sale listing

notifications, improving relevance by customizing the cadence and content of our notifications based on where users are in their home search, and ensuring our customers have a competitive edge by notifying them of new homes matching their preferences within minutes.”

To better support its objectives and a company-wide cloud migration effort, Redfin was interested in moving its email workloads from an internal data center to the cloud. “We wanted to save DevOps effort and time, reduce maintenance costs, increase infrastructure availability, and improve email deliverability metrics,” Misar says. Redfin faced a strict deadline to migrate its email service to the cloud. “We had one year to find a service and get it up and running. Otherwise, we would risk holding back the overall company cloud migration effort,” Misar says.

Solution | Moving Email from SMTP Servers to Amazon SES

Redfin had already begun migrating critical workloads, such as business analytics applications, onto Amazon Web Services (AWS). After meeting with AWS about email solutions, Redfin learned about the features and monitoring capabilities in [Amazon Simple Email Service](#) (Amazon SES), a cloud email tool that also reports on sender statistics and email deliverability. “We had compiled a list of requirements for our future email platform and evaluated a variety of email platforms,” says Misar. “We found Amazon SES to be the best match for our requirements.”

Working with AWS, Redfin created a plan for sending emails through Amazon SES instead of Redfin SMTP servers. After reviewing the full migration plan with the Amazon SES team, Redfin developers began implementing the design and sending emails through Amazon SES. Once the company confirmed that the systems worked as expected, it began sending larger volumes, gradually ramping up over a 45-day period. Throughout this process, Redfin closely monitored deliverability—including send, deliver, click, bounce, and complaint rates—to ensure email deliverability was not affected in the new system. “Our collaboration with the Amazon SES team helped our migration of millions of emails go smoothly,” says Misar.

Post migration, Redfin continues to use its Apache Kafka pipeline to process a large number of real estate listings to find matches and recommendations for users. Emails are then sent to users through Amazon SES, which also integrates with the company’s [Amazon Elastic Compute Cloud](#) (Amazon EC2) environment. The emails inform users about updates to homes for sale or for rent, such as price drops, open houses or new listings. Redfin also sends emails that recommend homes that a user may be interested in based on their home search journey. Redfin also uses [Amazon Simple Notification Service](#) (Amazon SNS) topics and [Amazon Simple Queue Service](#) (Amazon SQS) to process email bounces and complaints.



Outcome | Reliably Delivering Millions of Emails Daily

Using Amazon SES, Redfin sends millions of emails every day, with a peak send rate of over 1,000 emails per second. “In August 2023, we sent hundreds of millions of emails through Amazon SES, with a 99% deliverability rate,” says Misar. Redfin also has enhanced insights and alerting about its email deliverability and reputation across ISPs, including metrics to track delivery rates, bounces, and complaints. “Post migration, we were able to use the Amazon SES Virtual Deliverability Manager feature to break down this data by ISP. This has already helped us identify a deliverability issue with one large ISP, which was quickly resolved with the help of Amazon SES,” Misar says.

In addition, the Amazon SES feedback classification system provides accurate and reliable bounce and complaint notifications, features that improve Redfin’s email feedback handling system and email reputation. “Another large benefit is that we no longer need to maintain our own SMTP servers and can rely on the email expertise of Amazon SES,” adds Misar. “This reduces Redfin engineering costs, allowing for more engineering time to be spent on business growth.”

“The Amazon SES team were great partners. They made sure our plan was solid and were available to help us make our transition to the service a success,” Misar says. “They’ve also given us the opportunity to provide feedback and influence their product roadmap to accommodate our requirements.”

About Redfin

Redfin, based in Seattle, Washington, is a technology-powered real estate company that operates the leading real estate brokerage site in the US. Since its launch in 2006, the company has saved customers more than \$1.5 billion in commissions. Redfin serves more than 100 markets across the US and Canada and employs over 4,000 people.

AWS Services Used

Amazon SES

Amazon Simple Email Service (Amazon SES) lets you reach customers confidently without an on-premises Simple Mail Transfer Protocol (SMTP) email server using the Amazon SES API or SMTP interface.

[Learn more »](#)

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute portfolio with over 750 instances and choice of the latest processor, storage, networking, operating system, and more.



model to help you best match the needs of your workload.

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Amazon SNS

Amazon Simple Notification Service (Amazon SNS) sends notifications two ways, A2A and A2P. A2A provides high-throughput, push-based, many-to-many messaging between distributed systems, microservices, and event-driven serverless applications.

[Learn more »](#)

Amazon SQS

Amazon Simple Queue Service (Amazon SQS) lets you send, store, and receive messages between software components at any volume, without losing messages or requiring other services to be available.

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